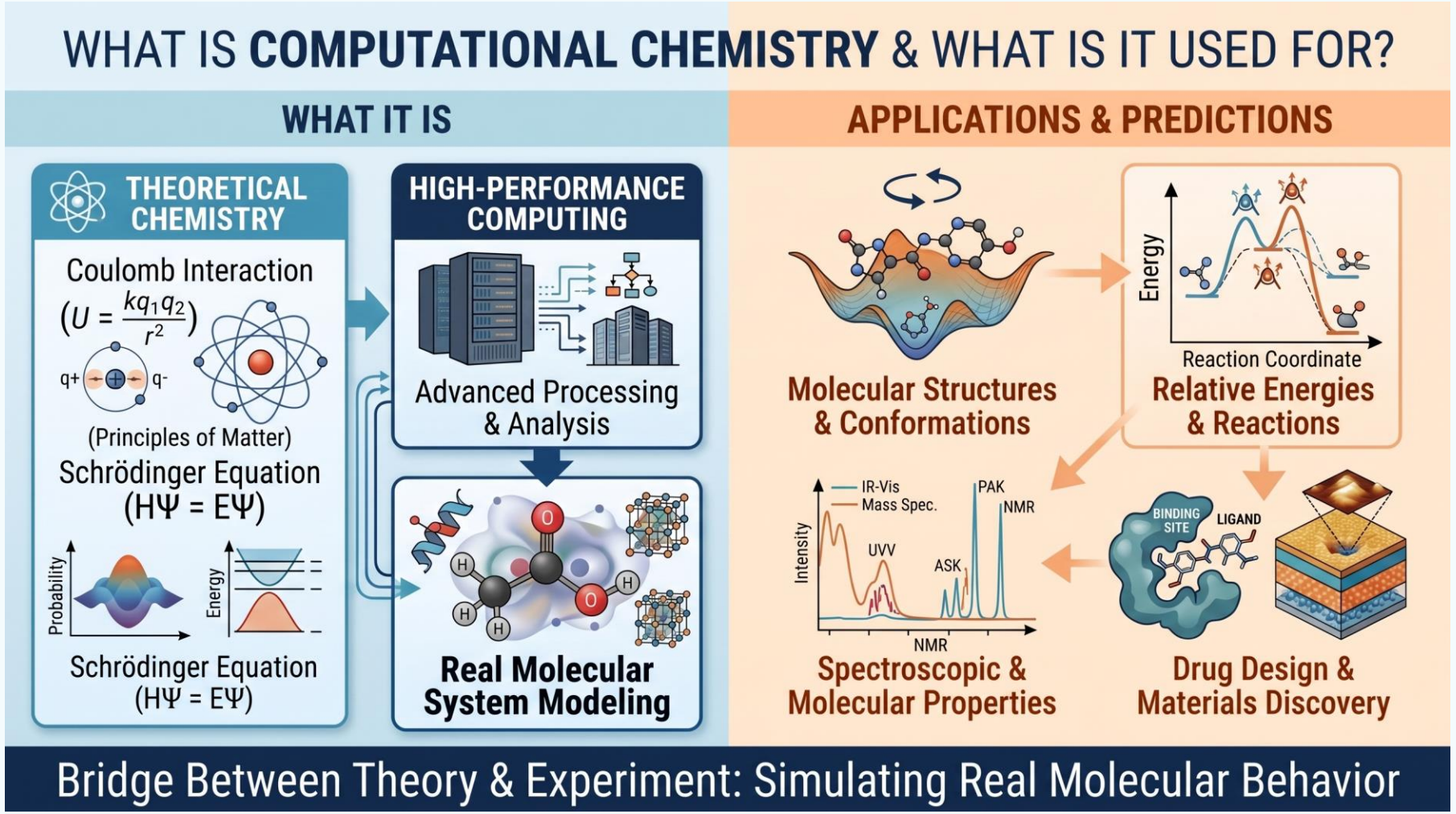
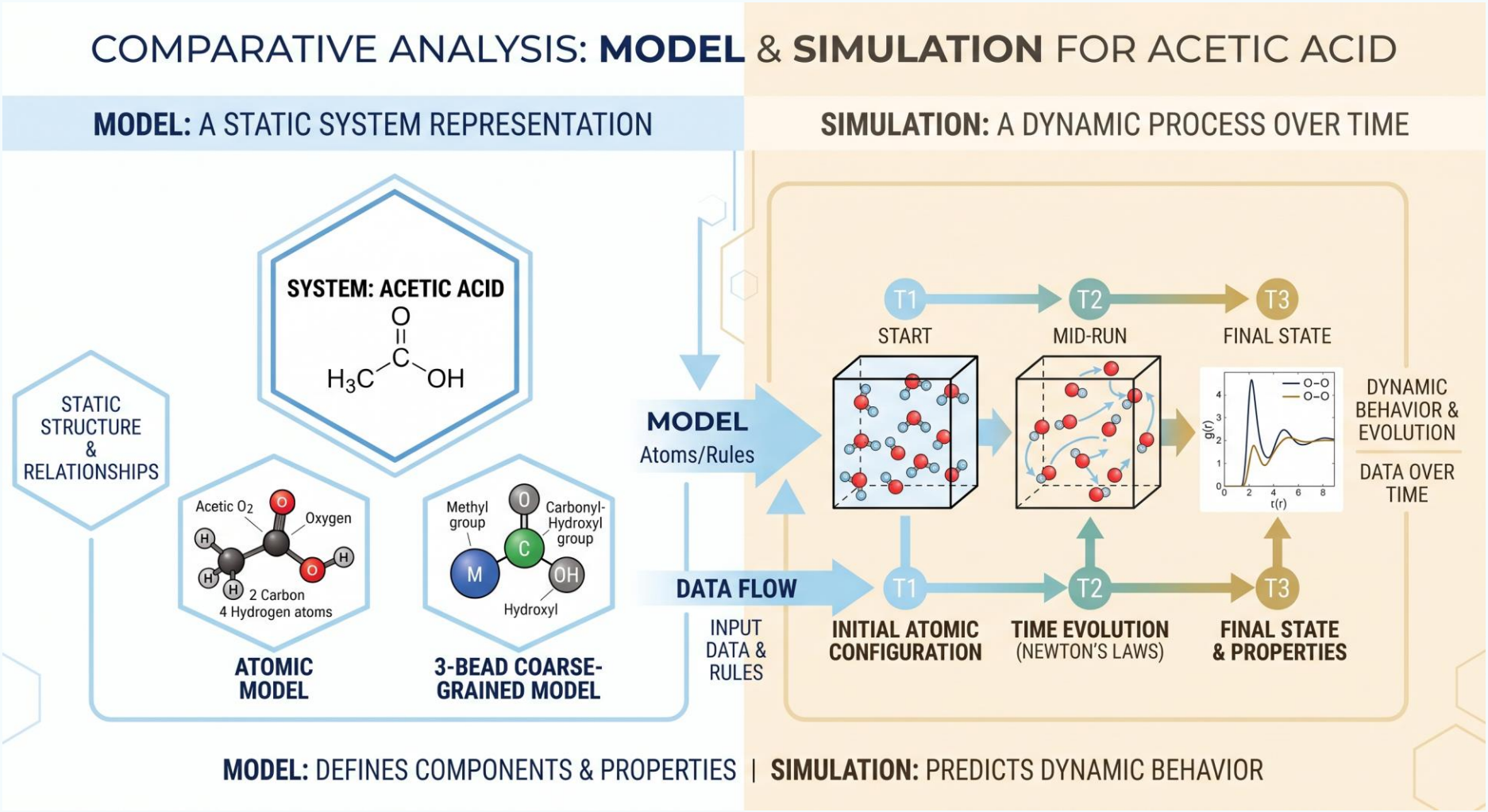
Introduction to Molecular Simulations

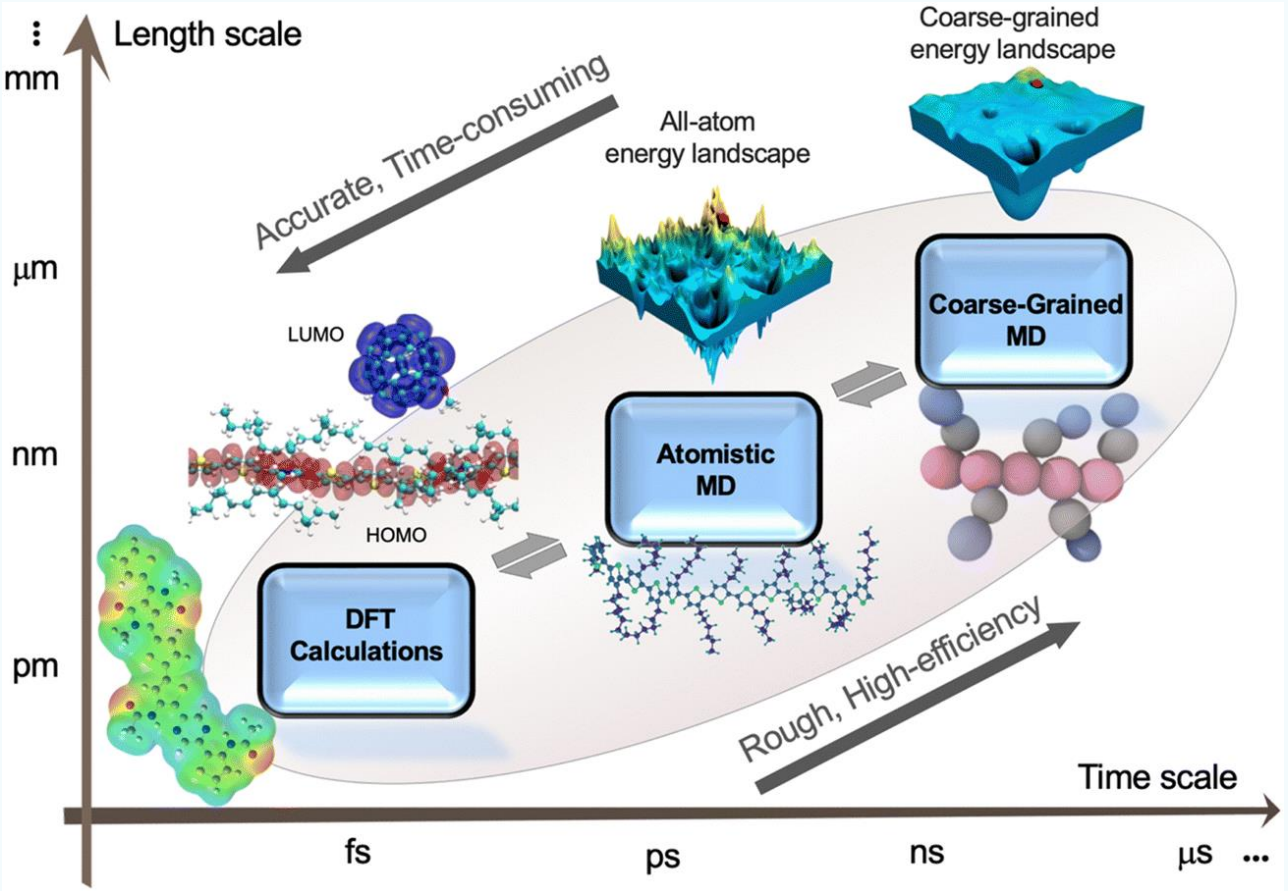
Alejandro H. Tanguma

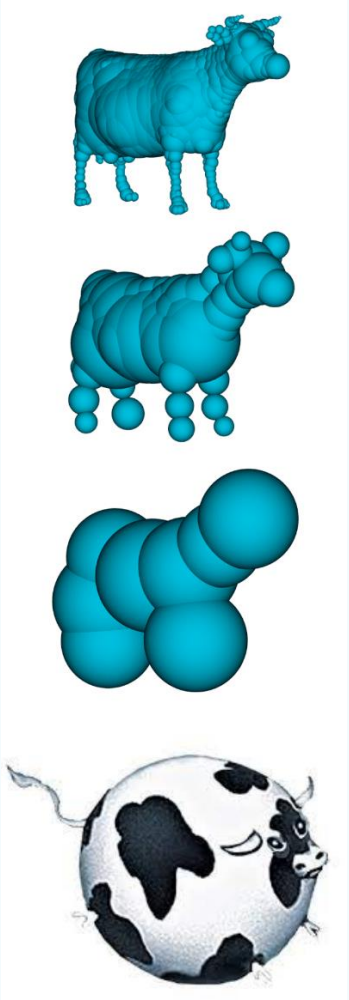
What is computational chemistry



Models and simulations

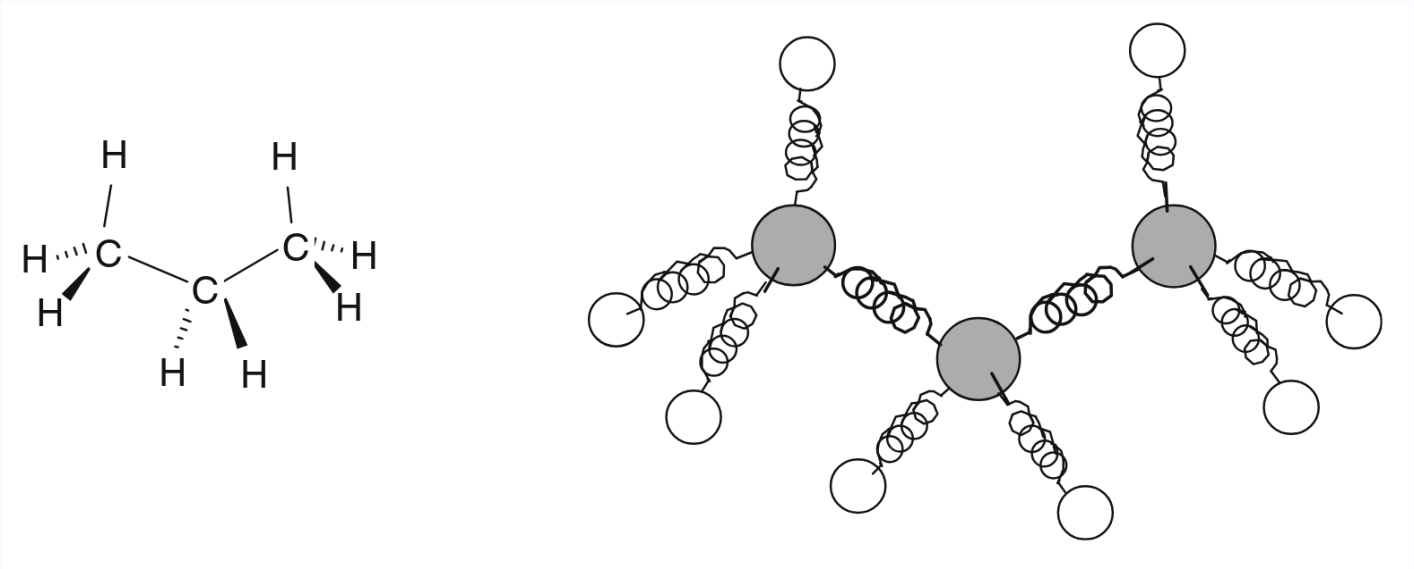






Tolba, Wang, Alesadi & Xia. Chem. Comm. 2024, 60, 11625-11641.

Molecular mechanics



E.G. Lewars. Computational Chemistry. 2016.

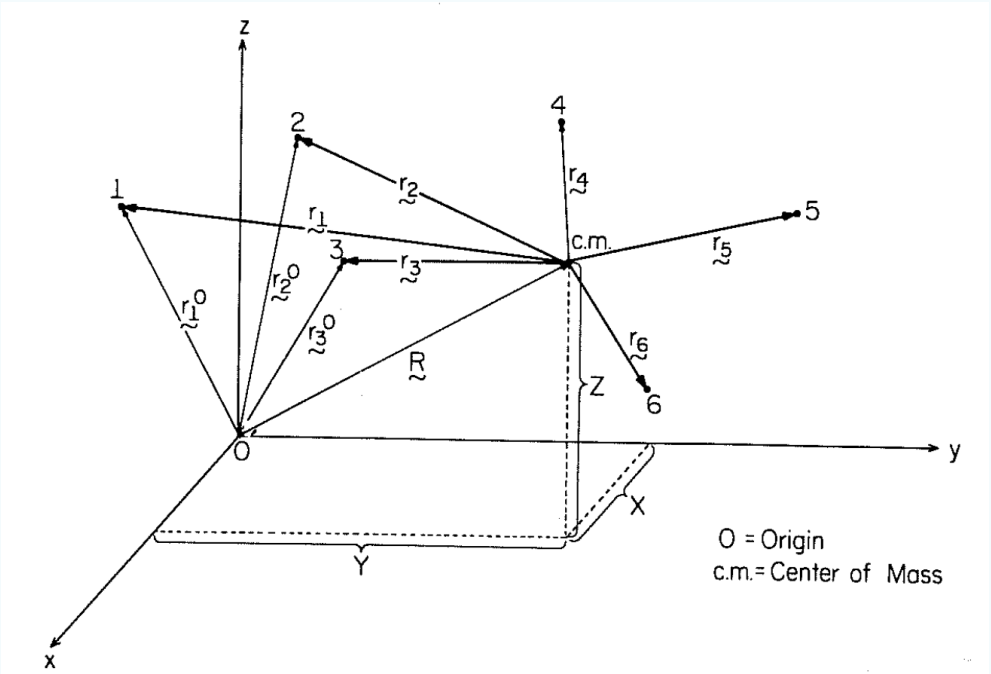


Figure 2-1 Cartesian coordinate axis system. 0 is the origin of the coordinate system chosen by convenience to be the laboratory frame and c.m. is the center of mass based in the many-particle frame. The r_i^0 and r_i vectors originate at the arbitrary origin and c.m., respectively, and terminate at the point masses designated by 1, 2, ... R is the vector from the arbitrary origin to the c.m. and X, Y, and Z are the Cartesian components of R.

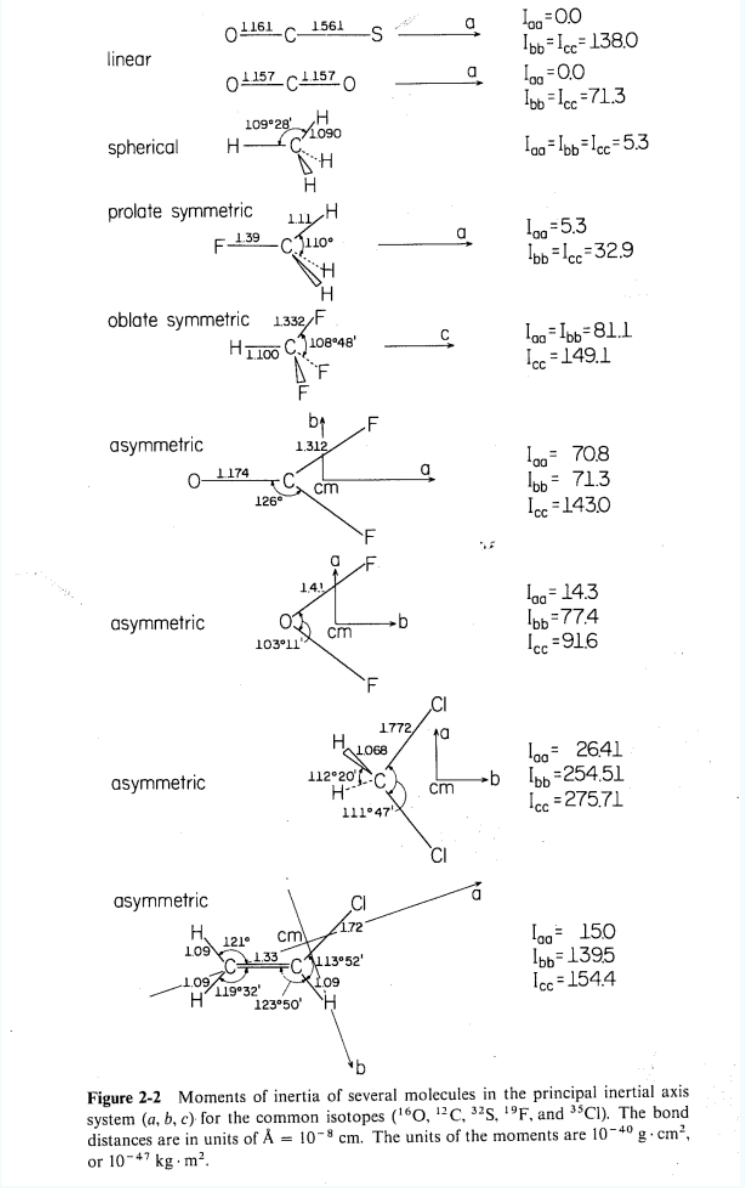
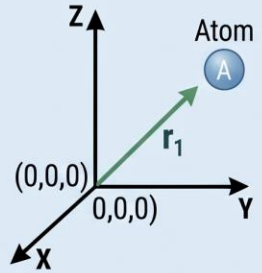


Figure 2-2 Moments of inertia of several molecules in the principal inertial axis system (a, b, c) for the common isotopes (^{16}O, ^{12}C, ^{32}S, ^{19}F, and ^{35}Cl). The bond distances are in units of Å $= 10^{-8}$ cm. The units of the moments are 10^{-40} g · cm^2, or 10^{-47} kg · m^2.

W. H. Flygare. Molecular structure and dynamics. 1978.

DEGREES OF FREEDOM (DOF) & PARTICLE COORDINATES

1 PARTICLE SYSTEM (N=1)

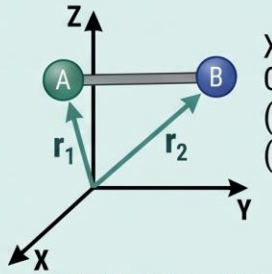


SYSTEM: 1 ATOM

3N = 3 DOFs

Coordinates: (x_1, y_1, z_1)

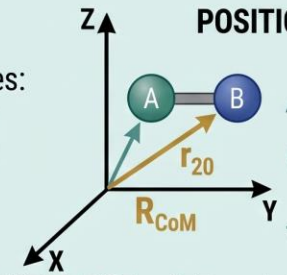
2 PARTICLE SYSTEM (N=2)



X, Y, Z Coordinates:
(x_1, y_1, z_1)
(x_2, y_2, z_2)

TOTAL DOF = 3N = 6

POSITION RELATIVE TO CoM

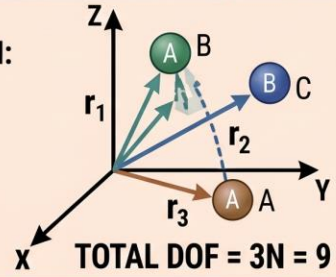


- **3** CoM Translational DOFs ($\mathbf{R}_{CoM}$)
- **2** Rotational DOFs (φ, θ)
- **1** Vibrational DOF (Bond, L)

TOTAL DOF = 6 (Translation + Rotation + Vibration)

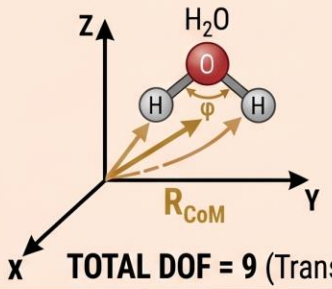
3 PARTICLE SYSTEM (N=3)

SYSTEM:
Lab
(Lab)



X, Y, Z Coordinates:
(x_1, y_1, z_1)
(x_2, y_2, z_2)
(x_3, y_3, z_3)

TOTAL DOF = 3N = 9



POSITION RELATIVE TO CoM

- **3** CoM Translational DOFs ($\mathbf{R}_{CoM}$)
- **3** Rotational DOFs (φ, θ, ψ - all independent)
- **3** Vibrational DOFs (2 Bonds, 1 Angle)

TOTAL DOF = 9 (Translation + Rotation + Vibration)

MOLECULAR MOTION TYPES & TYPICAL TIMESCALES (Small Molecules)

TYPES OF MOLECULAR MOTION

TRANSLATION

Movement of Center of Mass (CoM)

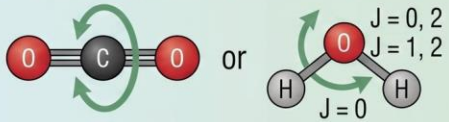


e.g., CO_2, of Center of Mass (CoM)

TRANSLATION
Molecule travels from **point A** to point **B**.

ROTATION

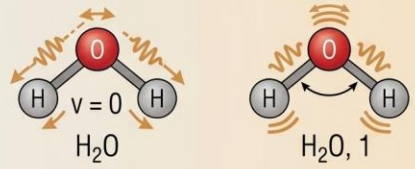
Rotation around Center of Mass



e.g., CO_2 e.g., H_2O

ROTATION
Molecule spins without displacing.

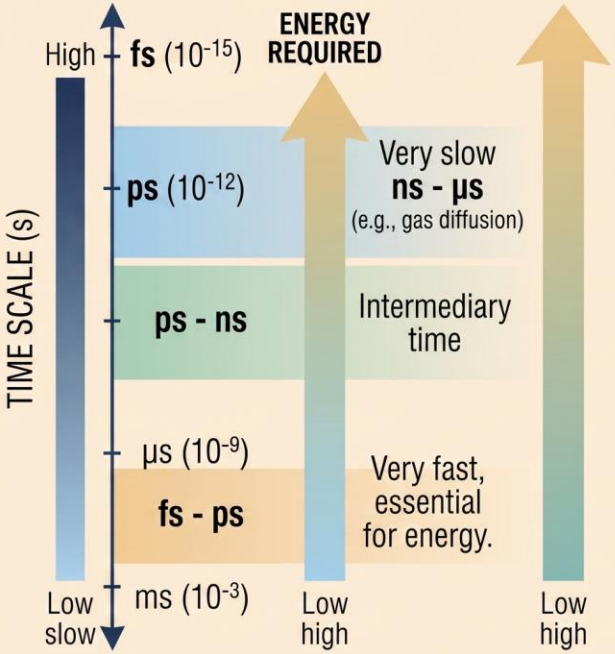
VIBRATION



H_2O H_2O, 1

Vibrational Modes: Stretches & Bends
Atoms oscillate around their equilibrium positions.

TYPICAL TIMESCALES



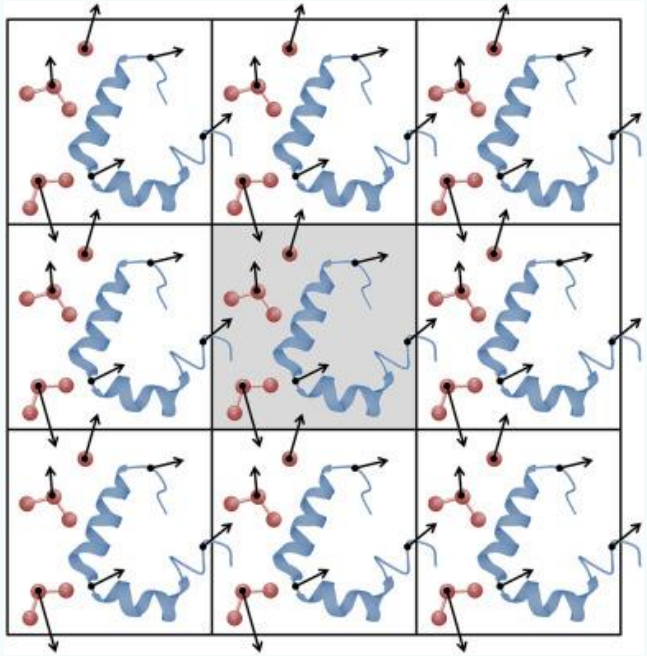
SUMMARY: TRANSLATION: Macro displacement | **ROTATION:** Fast spinning | **VIBRATION:** Ultrafast oscillation

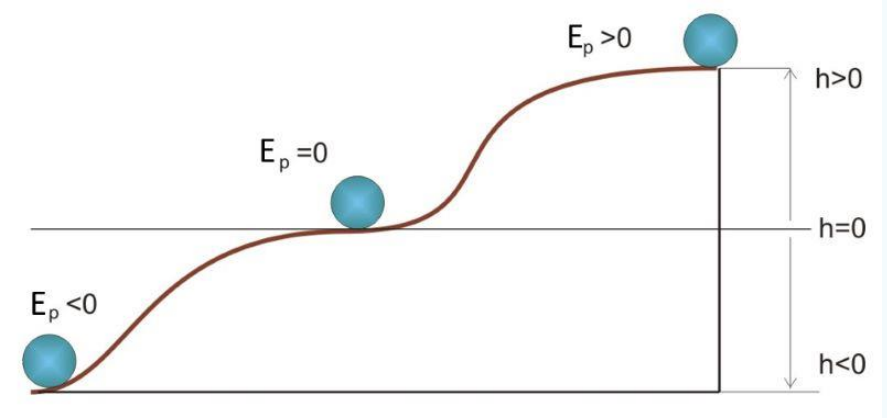
MD simulation

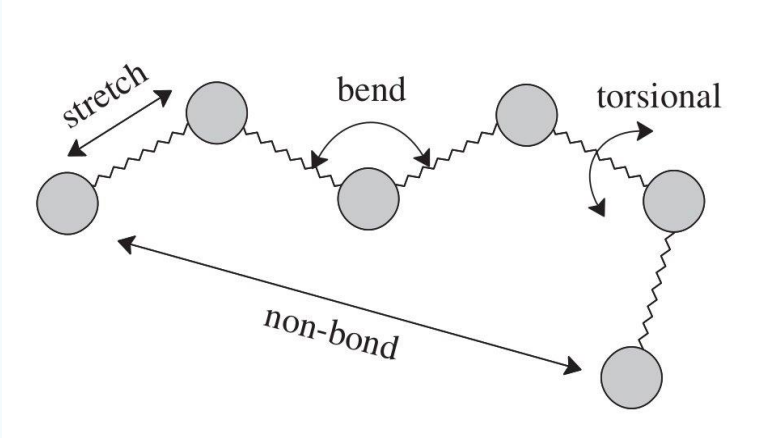
Description of the system: positions, masses, charges.

Initial conditions: velocities (temperature), boundary conditions.

Potential: interactions, external forces.

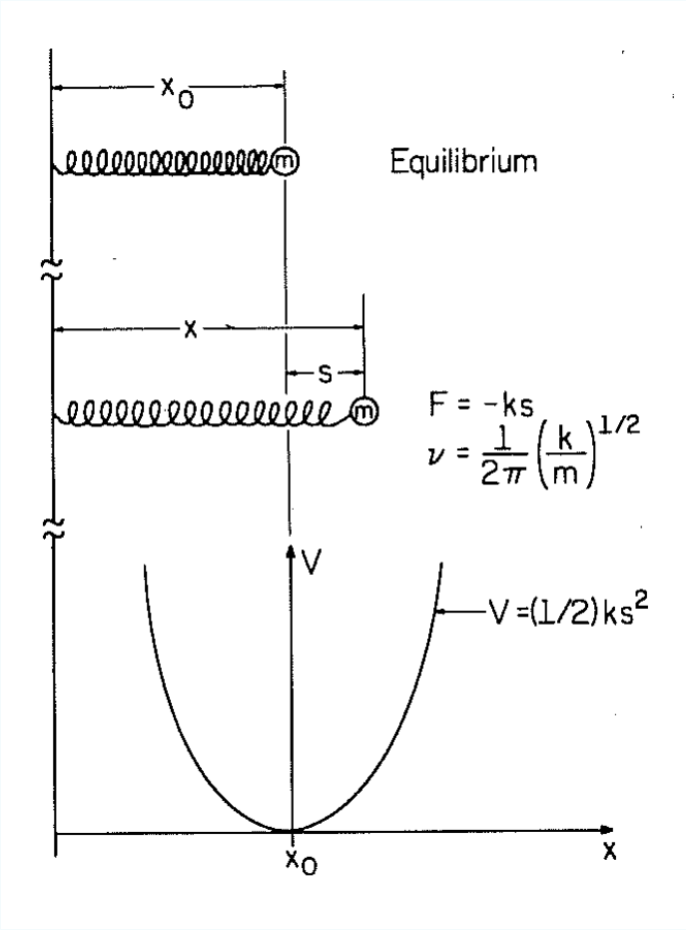






$$E = \sum_{bonds} E_{stretch} + \sum_{angles} E_{bend} + \sum_{dihedrals} E_{torsion} + \sum_{pairs} E_{nonbond}$$

F. Jensen. Introduction to computational chemistry. 2017.



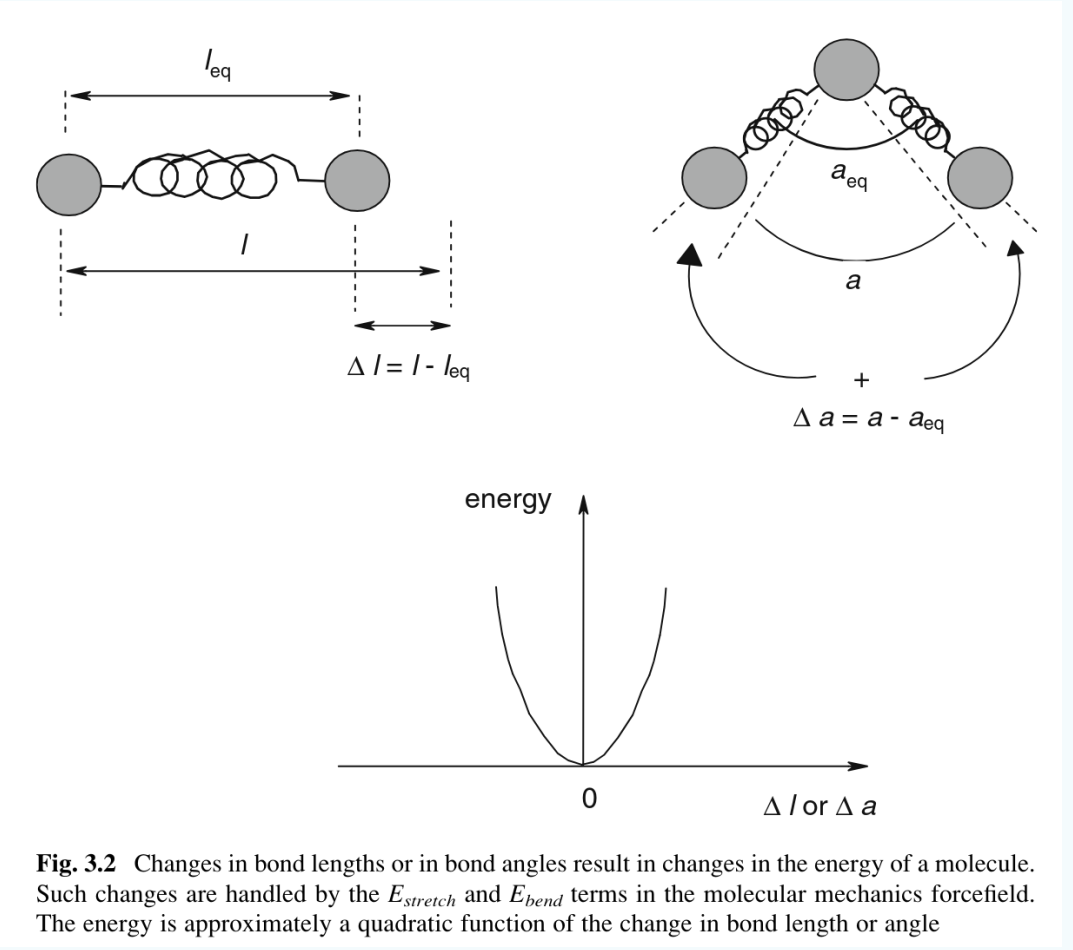
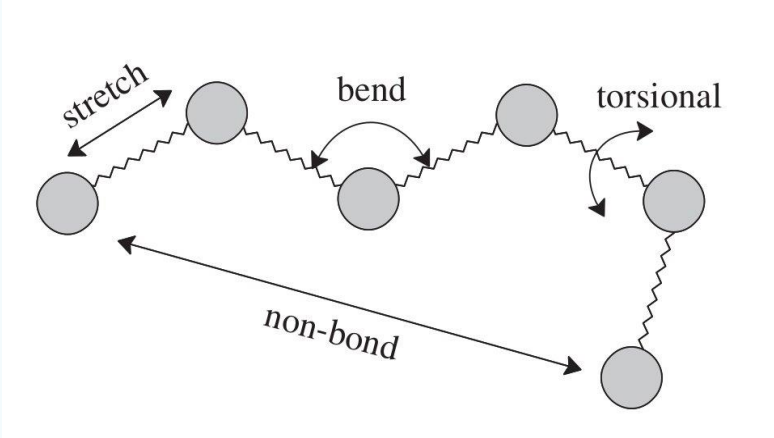


Fig. 3.2 Changes in bond lengths or in bond angles result in changes in the energy of a molecule. Such changes are handled by the $E_{stretch}$ and E_{bend} terms in the molecular mechanics forcefield. The energy is approximately a quadratic function of the change in bond length or angle



$$\mathcal{V} = \sum_{\text{bonds}} \tfrac{1}{2} k_r (r_{ij} - r_0)^2 + \sum_{\text{angles}} \tfrac{1}{2} k_\theta (\theta_{ijk} - \theta_0)^2$$

$$+ \sum_{\text{torsions}} \sum_{n} k_{\phi,n} [\cos(n\phi_{ijk\ell} + \delta_n) + 1] + \sum_{\substack{\text{non-bonded} \\ \text{pairs}}} \left[\frac{q_i q_j}{4\pi\epsilon_0 r_{ij}} + \frac{A_{ij}}{r_{ij}^{12}} - \frac{B_{ij}}{r_{ij}^{6}} \right].$$

https://github.com/TangumaAH/intro_MD